

Human Tracking in 3D Point Cloud Sequences

Detection, Tracking, and Motion Estimation

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1 Introduction

This report presents the design and implementation of a complete pipeline for processing, detecting, and tracking a human within timestamped 3D point cloud sequences stored as PCD files. The goal of the project is to compute a consistent human trajectory, maintain a stable identity across frames, and estimate instantaneous and smoothed velocity.

The pipeline includes:

- Loading and parsing raw binary PCD files.
- Preprocessing and voxel downsampling.
- Human detection using DBSCAN clustering.
- Multi-frame temporal tracking via a Kalman Filter.
- Track-to-detection association using the Hungarian Algorithm.
- Trajectory and speed computation over time.

This workflow mirrors the perception stack used in autonomous robots, mobile mapping systems, and advanced motion analytics.

2 Data and Preprocessing

The dataset of sequential PCD files represents a single LiDAR-like frame with coordinates and color encoded in binary. Each PCD file contains:

- Fields: `x`, `y`, `z`, `rgb`
- Binary data layout: 16 bytes per point
- Unorganized point cloud
(HEIGHT = 1)

Raw point clouds may contain redundant points, sensor noise, and non-uniform density. A voxel grid filter with voxel size 0.03 m is used to reduce noise, normalize density, improve clustering stability, and reduce computational load. Downsampling provides a cleaner shape of the human cluster, making detection more robust.

3 Human Detection using DBSCAN

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is applied to detect human in each frame. It groups points based on density rather than predefined cluster counts. The reason that it matches best for this task is that it does not require specifying the number of clusters and handles irregular cluster shapes. It also naturally rejects isolated noise points. and works well with sparse data typical of LiDAR scans. It uses two key parameters:

eps : maximum neighborhood radius

min_samples : minimum points needed for a dense region

The algorithm labels each point as: Core point (high local density), border point, or noise/outlier. Clusters are extracted from core point connectivity.

3.1 Centroid Extraction

After clustering, the human is assumed to be the largest cluster. Its centroid is computed as:

$$c = \frac{1}{N} \sum_{i=1}^N p_i$$

where $p_i = (x_i, y_i, z_i)$ are the points in the selected cluster.

This centroid becomes the measurement input to the tracking system.

4 Kalman Filter Tracking

A Kalman Filter is used to maintain a stable human identity and to smooth the noisy centroid measurements.

We model the human with a constant-velocity system. The 4D state vector is:

$$\mathbf{x} = [x, y, \dot{x}, \dot{y}]^\top$$

Prediction Step

The next state is obtained by applying the motion model A to the previously estimated state.

$$\mathbf{x}_{k|k-1} = A\mathbf{x}_{k-1|k-1}$$

The new state uncertainty is estimated by propagating the previous covariance through the motion model and adding process noise Q .

$$P_{k|k-1} = AP_{k-1|k-1}A^\top + Q$$

The Kalman gain determines how much the measurement should correct the predicted state.

$$K_k = P_{k|k-1} H^\top (H P_{k|k-1} H^\top + R)^{-1}$$

Update in the predicted state is derived by adding a correction based on the measurement residual weighted by the Kalman gain.

$$\mathbf{x}_{k|k} = \mathbf{x}_{k|k-1} + K_k(z_k - H\mathbf{x}_{k|k-1})$$

Updates in the state covariance reflects the reduced uncertainty after incorporating the measurement.

$$P_{k|k} = (I - K_k H) P_{k|k-1}$$

A is the motion model, H is the measurement model, and Q, R : process and measurement noise covariances.

The Kalman Filter provides Noise-reduced positions, smooth velocity estimates, and predictions for missing frames.

5 Data Association: Hungarian Algorithm

When multiple detections exist, we need to associate detected clusters with tracked identities. A cost matrix is built using Euclidean distance:

$$C_{ij} = \|t_i^{(\text{pred})} - d_j\|$$

The Hungarian Algorithm solves:

$$\min_A \sum_{i,j} C_{ij} A_{ij}$$

subject to one-to-one assignment constraints. This prevents ID switching and makes the system future-proof for multiple-person tracking.

6 Motion Analysis

6.1 Trajectory

The full sequence of estimated positions forms the human trajectory:

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

Figure 1 visualizes the human’s movement path as reconstructed from the tracked point-cloud centroids. Each point represents the estimated spatial position of the target at a given timestamp,

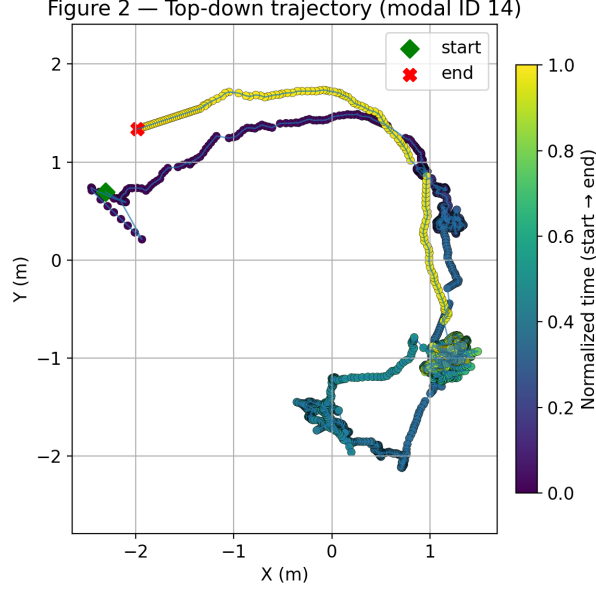


Figure 1: Reconstructed trajectory of the tracked human, obtained by plotting the sequence of Kalman-filter-estimated centroid positions across all frames.

smoothed through the Kalman filter to reduce measurement noise and compensate for missing or inconsistent detections. The resulting trajectory reveals the overall motion pattern, including direction, path curvature, and any positional drift caused by intermittent detection failures. This plot is crucial for verifying that the tracker maintains spatial coherence over time and for identifying sections where tracking uncertainty, occlusion, or segmentation artifacts may have influenced the estimated motion path.

6.2 Speed Over Time

Velocity magnitude is defined as:

$$v_k = \sqrt{\dot{x}_k^2 + \dot{y}_k^2}$$

The temporal evolution of the human’s estimated walking speed throughout the sequence is shown in Figure 2. The speed is derived from the magnitude of the velocity vector produced by the Kalman filter, which integrates measurements from noisy point-cloud detections while enforcing physically plausible motion through its prediction model. Variations in the curve correspond to natural changes in movement, temporary slowdowns, or brief instability caused by missed detections. Periods of flat or near-zero velocity often indicate occlusions or low-density frames in which the filter propagated the previous state without strong measurement updates. This plot provides a quantitative assessment of motion smoothness and highlights the filter’s effectiveness in mitigating noise from the underlying point-cloud data.

The temporal evolution of the human’s estimated walking speed throughout the sequence is reported in Figure 3. The speed is derived from the magnitude of the velocity vector produced by

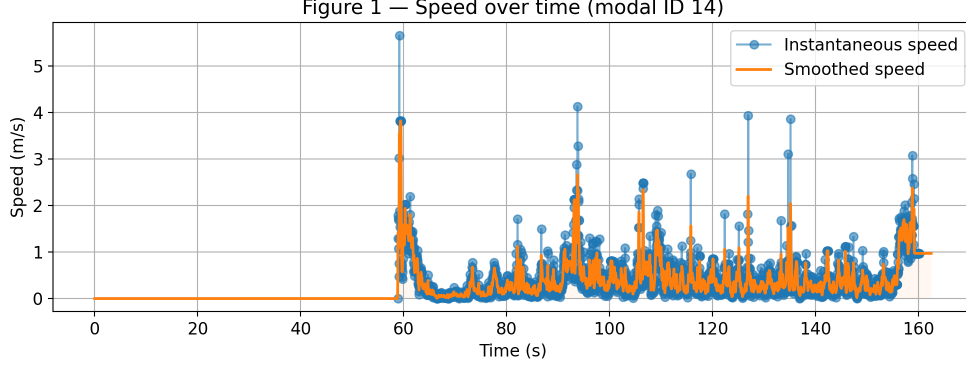


Figure 2: Estimated human speed (m/s) over time, computed from the Kalman-filter-smoothed velocity components extracted from the tracked point-cloud centroids.

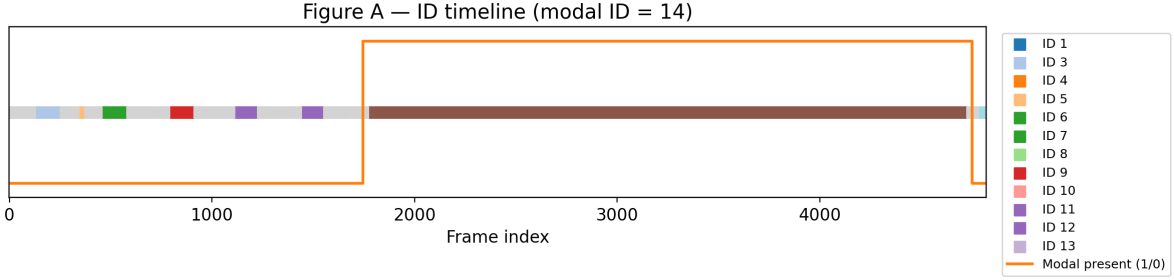


Figure 3: Estimated human speed (m/s) over time, computed from the Kalman-filter-smoothed velocity components extracted from the tracked point-cloud centroids.

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Figure 4 provides an overview of the stability and robustness of the tracking system by visualizing the aggregated ID-related statistics. The number of fragments indicates how often the tracker temporarily lost the main human track, typically due to occlusion or insufficient point density. The count of frames containing the modal ID reflects how consistently the tracker was able to maintain identity across the sequence. The presence of multiple distinct IDs highlights how often the tracker mistakenly assigned new labels, while the estimated ID switches provide an approximate measure of identity instability. Together, these metrics give a high-level understanding of the performance of the identity-assignment component of the pipeline.

To investigate how the number of detections varies over time, Figure 5 provides insight into the density, continuity, and consistency of the extracted clusters. Frames with zero detections typically correspond to cases where the human was partially or fully occluded, or the point cloud was too

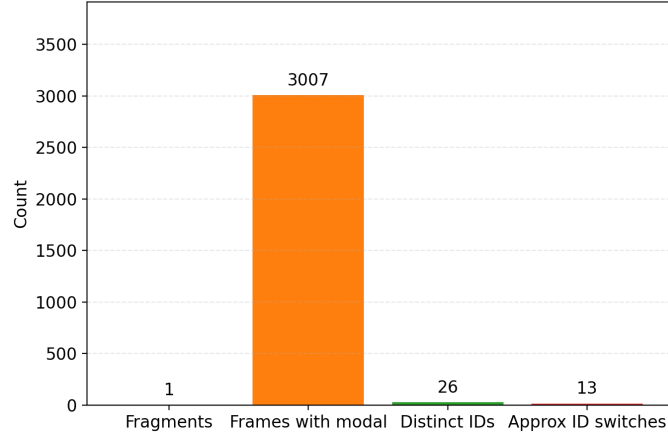


Figure 4: Summary of tracking performance metrics, including the number of track fragments, the number of frames in which the modal ID is present, the count of distinct IDs detected across the sequence, and the approximate number of ID switches.

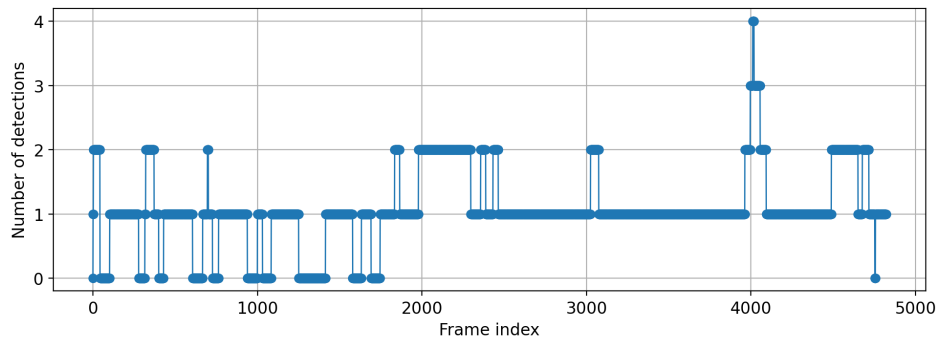


Figure 5: Number of human cluster detections per frame across the full sequence, visualized as a time series.

sparse for DBSCAN clustering to identify a valid cluster. Frames with more than one detection may indicate noise clusters, segmentation errors, or transient artifacts in the environment. Observing these fluctuations helps identify bottlenecks in the clustering stage of the pipeline and reveals how detection inconsistencies propagate into the tracking and velocity estimation steps.

7 Conclusion

This project demonstrates a complete and modular perception pipeline for human tracking using point cloud sequences:

- A custom binary PCD loader ensures compatibility with the dataset.
- Voxel grid downsampling improves data quality.
- DBSCAN detects the human robustly from raw point clouds.
- A Kalman Filter provides smooth, stable tracking over time.
- The Hungarian Algorithm ensures identity consistency.
- Final trajectory and speed plots summarize human motion.

The system mirrors real-world robotics and autonomous sensing architectures, and it can be extended to track multiple subjects, fuse additional sensor modalities, or integrate advanced deep-learning-based feature extractors.

8 Future Work and Further Analysis

Several extensions and improvements can be considered to enhance the accuracy, robustness, and interpretability of the tracking system. The most relevant directions for future work include the following:

- **Extended Motion Modelling:** The current Kalman Filter tracks only position and velocity, but future work could incorporate acceleration or full 3D kinematic models. This would improve prediction accuracy during rapid changes in motion, such as sudden stops or turns.
- **Richer Trajectory Analysis:** Additional motion metrics, such as trajectory curvature, turning behaviour, stability indices, or uncertainty evolution over time, could provide a deeper understanding of human movement patterns.
- **Improved Point-Cloud Preprocessing:** Techniques such as ground-plane removal using RANSAC, static-map subtraction, or statistical outlier filtering could significantly increase the reliability of human extraction from noisy point clouds.
- **Alternative Clustering Methods:** Beyond DBSCAN, algorithms such as HDBSCAN, Euclidean Cluster Extraction (PCL), or adaptive clustering methods could better handle varying point densities and more irregular human shapes.

- **Shape- and Height-Based Constraints:** Incorporating human height priors or bounding-box shape constraints can help eliminate false detections and enforce biological plausibility.
- **Advanced Tracking Filters:** Non-linear filters such as the Extended Kalman Filter (EKF), Unscented Kalman Filter (UKF), or Particle Filter may improve tracking under curved trajectories, non-linear dynamics, or high noise conditions.
- **Multi-Target Tracking:** For scenarios with multiple humans, frameworks such as Multiple Hypothesis Tracking (MHT) or Joint Probabilistic Data Association (JPDA) could be used to reduce identity switches and improve temporal consistency.
- **Enhanced Data Association:** Future systems could combine spatial information with appearance cues (e.g., colour) or apply gating strategies based on motion predictions to improve robustness during occlusions.
- **Adaptive DBSCAN Parameters:** Dynamic tuning of *eps* or *min_points* based on camera distance or local density could lead to more consistent clustering across the entire scanning range.
- **Uncertainty and Noise Analysis:** Evaluating how noise propagates through detection and tracking stages, as well as quantifying uncertainty during occlusions or sparse point clouds, would help refine model parameters.
- **Quantitative Evaluation:** If ground-truth trajectories become available, performance can be evaluated using positional error, velocity error, and trajectory deviation metrics. Ablation studies could also identify the impact of key parameters such as voxel size, DBSCAN thresholds, or Kalman noise matrices.
- **Real-Time Implementation:** Optimising or parallelising the pipeline, integrating it into ROS, or using GPU acceleration would allow deployment in real-time systems. Streaming visualisation tools could further improve usability.
- **Sensor Fusion:** Combining LiDAR point clouds with complementary sensors such as RGB-D cameras or IMUs could greatly improve robustness in challenging environments.
- **Improved Visualisation:** Future work may include displaying uncertainty ellipses, motion vectors, bounding boxes, or multiple tracked paths for clearer interpretation of the tracking results.

Overall, these extensions represent promising opportunities to strengthen the tracking pipeline, improve the accuracy of the extracted trajectories, and enhance the system’s applicability to real-world scenarios.

9 Appendix

The tracker uses a 6D state transition matrix:

$$\begin{pmatrix} x \\ y \\ z \\ v_x \\ v_y \\ v_z \end{pmatrix} \quad (1)$$

$$x_0 = \begin{pmatrix} p_0 \\ 0_3 \end{pmatrix} \quad (2)$$

$$P_0 = 0.1I_6 \quad (3)$$

The constant-velocity state transition matrix:

$$F(\Delta t) = \begin{pmatrix} I_3 & \Delta t I_3 \\ 0_3 & I_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 & \Delta t & 0 \\ 0 & 0 & 1 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \quad (4)$$

Process noise is modeled as:

$$Q = qI_6, \quad q = 10^{-2} \quad (5)$$

Prediction equations:

$$\begin{aligned} x_{k|k-1} &= F(\Delta t)x_{k-1|k-1} \\ P_{k|k-1} &= F(\Delta t)P_{k-1|k-1}F(\Delta t)^T + Q \end{aligned} \quad (6)$$

Measurement matrix selects position components:

$$H = \begin{pmatrix} I_3 & 0_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix} \quad (7)$$

measurement noise:

$$R = rI_3 \quad r = 10^{-2} \quad (8)$$

Residual/ innovation in Kalman filter:

$$y_k = z_k - Hx_{k|k-1} \quad (9)$$

$x_{k|k-1}$ filter's prediction

$Hx_{k|k-1}$ the predicted measurement

z_k : the actual measurement from the sensor

innovation = measurement-prediction

The Kalman update step uses the innovation to adjust the predicted state:

$$x_{k|k} = x_{k|k-1} + K_k y_k \quad (10)$$

Innovation covariance combines uncertainty from both prediction and measurement:

$$S_k = HP_{k|k-1}H^T + R \quad (11)$$

Kalman gain:

$$K_k = P_{k|k-1}H^TS_k^{-1} \quad (12)$$

state update:

$$\begin{aligned} x_{k|k} &= x_{k|k-1} + K_k y_k \\ x_{k|k} &= x_{k|k-1} + K_k(z_k - Hx_{k|k-1}) \end{aligned} \quad (13)$$

the Kalman gain controls how strongly new measurements pull the estimate.

Covariance update:

$$P_{k|k} = (I_6 - K_k H)P_{k|k-1} \quad (14)$$